



Mechanical reinforcement of natural rubber (NR)–ethylene-propylene-diene monomer (EPDM) blends via BMI (1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide)- and TESPT (bis[3-(triethoxysilyl)propyl] tetrasulfide)-functionalized carbon nanotubes

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Abstract

Carbon nanotubes (CNTs), particularly when surface-functionalized, are highly effective reinforcements for natural rubber/ethylene-propylene-diene monomer (NR–EPDM) composites, enhancing both mechanical strength and swelling resistance. In this study, unmodified CNTs and CNTs functionalized with 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (BMI) and bis[3-(triethoxysilyl)propyl] tetrasulfide (TESPT) were incorporated into the NR–EPDM matrix at loadings up to 8 parts per hundred rubber (phr) using conventional melt mixing. The curing characteristics were analyzed using an oscillating disc rheometer, and detailed evaluations of mechanical, swelling, and compression properties were performed. Unmodified CNTs increased cure torque and reduced cure time, resulting in improvements in tensile strength, 100% modulus, and tear strength by 25%, 26%, and 22%, respectively, with optimum performance at 5 phr. However, higher filler loadings caused CNT agglomeration, which reduced the tensile strength and modulus. Composites containing BMI-functionalized CNTs exhibited superior reinforcement, achieving maximum increases of 39% in tensile strength and 44% in modulus at 5 phr, while tear strength improved by 29% compared to the unfilled blend. Composites reinforced with TESPT-functionalized CNTs showed the most significant enhancements, with tensile strength, modulus, and tear strength increasing by 52%, 56%, and 37%, respectively. These composites also displayed moderate reductions in elongation at break and rebound resilience. TESPT modification provided excellent swelling resistance, with peak performance observed at 5 phr, followed by a slight decline at higher filler concentrations due to filler clustering. Overall, TESPT-functionalized CNTs achieved better dispersion and stronger interfacial bonding within the NR–EPDM matrix, leading to remarkable improvements in mechanical integrity, crosslink density, and solvent resistance.

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